Info

Welcome!

This app lets you explore how sleep, stress, study behavior, and academic performance are related. Each tab shows one of the datasets we used in our project, and you can choose your own variables to plot on the x– and y–axes. We decided to make a separate tab for the histogram distribution because we though it tied directly into answering the major question that was lingering in our heads.

Sleep Dataset Tab

This tab uses the Sleep Health and Lifestyle dataset. Here you're mainly looking at sleep duration, sleep quality, and stress level, along with age and gender.

- Use the Plot-X and Plot-Y dropdowns to choose any two numeric variables from the sleep dataset (for example,
  - `sleep_duration` (hours of sleep)
  - `sleep_quality` (self-reported quality)
  - `stress_level` (self-reported stress)

As you change x and y, the correlation summary at the top updates to tell you how strong and in what direction the relationship is. For example:

- If you set X = `sleep_duration` and Y = `stress_level`, a negative correlation means that people who sleep more tend to report lower stress.
- If you set X = `sleep_duration` and Y = `sleep_quality`, a positive correlation suggests that more sleep is associated with better sleep quality.

Use this tab to explore our first research question: *What do general sleep patterns look like, and how do sleep duration, quality, and stress relate to each other?*

Additional Functionalities

- The slider under “Bounds for X” lets you zoom in on a specific range of the x-axis. For example, you can focus only on people sleeping between 5 and 8 hours.
- The point size slider changes how large each dot is.
- The Add best fit line box overlays a straight regression line (linear relationship).
- The Add best fit curve box overlays a smooth curve (nonlinear trend).
- The Remove background box switches to a more minimal theme.

Student Dataset Tab

This tab uses the Student Performance dataset. Key variables here include:

- `performance_index` - overall performance score
- `previous_scores` - earlier exam or course performance
- `hours_studied` - study time
- `sleep_hours` - hours of sleep for each student

Again, you can choose any two numeric variables as X and Y:

- X = `sleep_hours`, Y = `performance_index`
  - Shows how nightly sleep relates to current academic performance. A positive slope suggests that students who sleep more tend to do a bit better.
- X = `hours_studied`, Y = `performance_index`
  - Shows how study time relates to performance. A positive slope suggests “more study, slightly higher performance.”
- X = `previous_scores`, Y = `performance_index`
  - Shows how past performance predicts current performance (usually a strong positive relationship).
- Use the bounds slider to restrict the x-axis (for example, only students studying less than 20 hours), and use the line/curve options to see trend patterns more clearly.
- The correlation summary helps compare relationships:
  - A strong positive correlation (e.g., between previous scores and performance index) indicates a strong academic channel.
  - A weak but positive correlation (e.g., between sleep_hours and performance_index) shows that sleep helps, but is not the only factor.

This tab helps explore: *How are sleep and academic performance related, and how does sleep compare to study time and prior performance?*

Factors Dataset Tab

This tab uses the Student Performance Factors dataset. Here, important variables include:

- `study_hours` - time spent studying
- `sleep_hours` - sleep habits
- `score` - exam score or performance measure

Try combinations like:

- X = `study_hours`, Y = `score`
  - Shows how exam scores respond to increased study. The line tells you whether more study is generally helpful.
- X = `sleep_hours`, Y = `score`
  - Shows whether students with more sleep tend to get higher scores.
- As in the other tabs, the correlation summary provides a quick read:
  - A moderate positive correlation between `study_hours` and `score` would support the idea that more study leads to better outcomes.
  - A small positive correlation between `sleep_hours` and `score` would suggest that sleep has a gentle but beneficial association with performance.

This tab focuses on the behavioral side of our question: *How do sleep and study habits connect to exam performance?*

Joined Dataset Tab

The Joined Dataset tab uses a combined dataset that aggregates all three sources by rounded sleep hours. For each sleep-hour value, we summarize:

- Average performance index and previous scores (from the student dataset)
- Average exam scores and study hours (from the factors dataset)
- Average sleep quality and stress levels (from the sleep dataset)

You can choose any two summary variables as X and Y, such as:

- X = `sleep_duration` or rounded `sleep_hours`, Y = `avg_performance`
  - Shows how average sleep relates to average academic performance across all datasets.
- X = `sleep_duration`, Y = `avg_stress_level`
  - Shows how stress changes as sleep duration increases.

- X = `sleep_duration`, Y = `avg_sleep_quality`
  - Shows how average sleep quality varies across sleep durations.
- X = `avg_study_hours`, Y = `avg_score`
  - Combines study behavior and performance into a higher-level view.

This tab is designed to support our final story: *Students who sleep within the recommended range tend to show higher sleep quality, lower stress, and slightly better academic performance than short sleepers.* It lets you see how patterns line up when we combine all datasets into one "big picture."